
ADSL is one of the strongest contenders in the race to deliver broadband

Let's begin by asking a question. What's the maximum data transfer speed you can get over your home telephone line?

Being used to 56K modems, you probably thought the maximum speed you could get from your telephone line was 56 kilobits of data per second. Now read this: on that same phone line, you can actually get data rates reaching 8,192 kilobits per second. And we're not talking cable or ISDN here.

A landline telephone is today as commonplace in the average household as the TV or fridge. For most of us, the landline doubles up as our only window to the Wide World of the Web. Well, right now, there are quite a few competing technologies that offer connectivity and bandwidth solutions: Leased Lines, ISDN, Cable, Satellite links, LMDS (Local Multipoint Distribution Systems) and 3G, or Third Generation Wireless, are differing technologies that try to address the increasing demand for bandwidth. This demand has spilled over from large enterprises to small and medium businesses and increasingly, homes.

Since its inception in 1989 as an attempt at delivering streamed video

over copper phone lines, DSL, or Digital Subscriber Line, has branched into several similar, competing technologies. Combined, this family of technologies is known as xDSL. Asymmetric Digital Subscriber Line, or ADSL, is by far the most important member of this extended brotherhood.

Cable is the most widely deployed broadband solution in the United States, and also in India; However, DSL does have its advantages. You get a dedicated connection instead of a shared one, and surfing speeds therefore don't vary depending on the time of day.

Peeling off the insulation

First things first: DSL is a modem technology that uses existing twisted-pair telephone lines to digitally transport high-bandwidth data, such as multimedia or video, to subscribers. You can use your phone while being simultaneously connected to the Internet: The process entails a technique known as multiplexing (see box 'Multiplexers and Multiplexing' on page 99) to pack the two kinds of data and channel them through the existing telephone network. The two modems, one at the central office of the telecom

Turbocharging the tele-

